

Retro Stores

If you are getting bored of looking at those graphic-laden interfaces of play store and app store, then try out this retro store link to refresh. <http://dgit.in/RetroStore>



Blockchain tech

IBM has revealed its plans to fully commercialize the blockchain tech for usage beyond Bitcoins and money transfer. <http://dgit.in/ChainTech>

Geek Life

Where sky's the binary limit

Supercomputers are every nation's gateway to innovations, and are pivotal for R&D. So what's going on with supercomputing in India?

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Supercomputers are using their high computational speeds and efficiency to help many sectors in our country, such as scientific research, weather forecasting, missile simulation, understanding the creation of the universe, creating life saving pharmaceutical drugs and much more. Now, what really makes a Supercomputer "super" is a concept called parallel computing.

Let's take an example of painting a rainbow on a wall: serial processing is like a single man painting one colour at a time, and the time taken will depend on the speed at which the person draws each colour. Now, in a parallel processing system, with, say, seven people available to paint the rainbow, each person paints one colour, and the time taken is much less. Basically, parallel processing involves the breaking up of tasks into smaller tasks

that can be processed in parallel. The end result is obtained by combining outputs from each processor.

What a Supercomputer does?

Any gamer knows that adding a faster CPU, GPU and RAM means a better gaming experience, but there are limits to how well a single processor can handle loads.

A supercomputer consists of an array of processors and GPUs stacked together to perform high computational tasks.



An old supercomputer from the 1960s

These stacked processors are capable of very fast parallel processing. However, parallel computing does have some drawbacks, as combining outputs from thousands of processor requires sophisticated protocols that can arrange these thousands of outputs in the correct order. Some Supercomputers are not located in a single room, and are actually a collection of smaller supercomputers that are spread across large geographical distances. Weather forecasting needs such a setup.

A supercomputer can be built by stacking computer processors in a giant box and interconnecting them to work on a complex task through parallel processing. Such an arrangement is called a Cluster supercomputer. Here, each individual computer in the cluster is called a node. Another type is Grid supercomputers, where the processors are stacked in pretty much the same way as they are for a Cluster setup, but in this case can be located across the globe and connected via WANs or the internet.

Supercomputers in India

India is home to some of the top supercomputers in the world. We use supercomputers to conduct various scientific research, make meteorological predictions, for aerodynamic modelling, bioinformatics, and also simulating nuclear missile tests.

Unlike regular computers, where you measure performance in GHz, supercomputers are measured based on how many FLOPS (floating point operations per second) they are capable of. Think of a Flop as a way to calculate a number to a decimal place.

Top 5 supercomputers in India

India has 11 of the top 500 supercomputer in the world. Here are the top 5 of those mammoth supercomputers that serve our nation.

SahasraT (Cray XC40)

SahasraT supercomputer is located at Supercomputer Education and Research Centre (SERC) facility at Indian Institute of Science in Bangalore. SERC is India's state-of-the-art research facility for high